

Authority Diffusion Before Founders Leave Open Source Projects: A Unified-Corpus, Noise-Floor-Adjudicated Test

Abstract

Open-source projects that lose their sole founder-maintainer sometimes recover and sometimes stall, and prior work (Avelino et al., ESEM 2019) predicts this outcome from *snapshot* covariates measured at the moment of departure – project size, developer count, commit volume – with small reported effect sizes. We test whether the *trajectory* leading up to departure carries a stronger signal: specifically, whether authority is already diffusing away from the founder in the months before a Truck-Factor-Developer-Detachment (TFDD) event. This is a second, methodologically strengthened pass at that question: we mine a single unified corpus of founder-only TFDD events (superseding two disjoint corpora from a prior iteration), substantially raise the resolution of a within-repository placebo control, and add a new window-boundary-noise control that isolates measurement noise from signal – a validation step the underlying Degree-of-Authorship pipeline had never received, since its authors validated it only at yearly full-history snapshots, not on the sub-year windows a trajectory measure requires. All three pre-registered success criteria again fail, but now for three distinct and diagnosable reasons rather than a uniform sample-size ceiling: the regression runs to completion and the diffusion covariate falls well short of significance and of the snapshot covariates’ effect size; the matched-pairs test remains structurally untestable, now because of a genuine floor effect rather than ambiguous small counts; and the placebo test’s point estimate is individually striking, but a purpose-built noise-floor control shows it is not distinguishable from ordinary within-repository measurement noise. A supplementary bootstrap and manual identity-resolution audit further bounds two data-quality risks the prior iteration could only describe qualitatively. We again report a negative result, and we now localize its cause precisely: not corpus size alone, but a genuinely noisy sub-year measurement whose noise floor this corpus cannot yet be shown to exceed.

1 Introduction

Single-founder open-source projects are a recognized failure mode in software engineering: a project with Truck Factor 1 – one developer whose departure would immediately strand a majority of the codebase – can vanish the day that developer stops committing [Avelino et al., 2016]. Avelino et al. formalized this event as a Truck-Factor-Developer-Detachment (TFDD) and, in a follow-up study of 1,932 GitHub projects, found that 315 (16%) experienced one, of which only 128 (41%) survived it [Avelino et al., 2019]. Understanding which departures a project survives is directly actionable: it tells maintainers, funders, and platform operators which projects are worth a targeted succession intervention before the founder leaves, rather than after.

This question matters at scale. Foundation-scale audits of package ecosystems put single-maintainer exposure in the tens of percent of actively used packages [Jabrayilzade et al., 2022, Coelho and Valente, 2017], and turnover studies across RubyGems, npm, and Rust show departure is a recurring, not rare, event [Constantinou and Mens, 2017, Fan et al., 2025]. A cheap, well-calibrated early-warning signal – computed from public git history alone, with no access to private

roadmaps or contributor sentiment – would let a platform or foundation triage which projects to approach before, rather than after, the truck factor drops to zero.

The obvious signal is hard to extract cleanly. Avelino et al.’s own predictor is a *snapshot*: project size, developer count, and commit volume measured at the moment of the TFDD, with reported effect sizes in the small range (Cohen’s d 0.13–0.26) [Avelino et al., 2019]. A snapshot, by construction, cannot see whether the project was already quietly redistributing ownership before the founder left, or whether the founder’s exit came as a genuine surprise with no prior diffusion at all. Testing a *trajectory* signal instead requires recomputing the Degree-of-Authorship (DOA) pipeline not once at a fixed snapshot but continuously over a 6–12 month pre-departure window – a computation the original authors never validated, since their own use of it was confined to yearly full-history snapshots.

A first pass at this question, reported in a prior iteration of this work, tested a 30-event, independently-mined corpus and found all three pre-registered criteria failing for a single, uniform reason: too few founder-only TFDD events to give the regression and matched-pairs tests any usable within-corpus outcome variance, and a placebo test whose true-window correlation sat unremarkably at the 40th percentile of a null built from only 25 draws. That result left three concrete gaps: two disjoint corpora (a 32-repository dataset artifact and an independently re-mined 62-candidate experiment corpus) that were never reconciled, a placebo null too coarse to resolve percentiles finer than 4%, and no test of whether the sub-year window computation itself – rather than the underlying mechanism – could manufacture a spurious null or signal. This paper closes all three. We mine a single unified corpus at larger scale, raise the placebo draw count twelvefold (25 to 300), and add a window-boundary-noise control: we recompute the same diffusion covariate on repeated stable-period (non-departure) windows in a held-out set of repositories and use the resulting variance as a floor against which to judge whether the true pre-departure window’s effect is a genuine signal or ordinary window-recomputation noise. We also add a manual identity-resolution audit and bootstrap confidence intervals on the snapshot-covariate effect sizes, closing two data-quality gaps the prior iteration flagged but did not measure.

The result is a negative one, as before, but for a materially different and more diagnostic reason. With the corpus provenance unified and a working regression, the diffusion covariate (count of new non-founder file owners accruing authority pre-departure) is now testable and fails cleanly: its BH-FDR-adjusted p -value is 0.84, and its standardized effect size (0.13) trails the mean of the snapshot control covariates (0.34) by more than a factor of two. The matched-pairs test remains untestable, now for a sharper reason than before – 28 of 32 repositories fall in the high-diffusion stratum and zero in the low-diffusion stratum, a genuine floor/ceiling effect rather than an artifact of small-cell binary outcomes. The placebo test, finally well-resolved, places the true window at the 3rd percentile of its null – an extreme result that would ordinarily read as a real, if inverted, signal. The window-boundary-noise control we add specifically to adjudicate results like this one shows that stable-period windows in held-out repositories already exhibit a variance floor (mean 11.74, SD 3.43) far exceeding the placebo null’s own spread (SD 0.18); the true window’s effect sits within roughly 1.4 boundary-noise standard deviations of zero, short of the 2-SD threshold we pre-registered as the bar for treating an extreme placebo percentile as a genuine signal rather than measurement noise. We report this as a negative result and, unlike the prior iteration, one whose binding constraint is not simply “too few repositories” but a specific, now-measured noise floor in the sub-year DOA computation itself.

Summary of Contributions

- A unified, single-provenance corpus of 32 founder-only TFDD events mined from 216 screened GitHub candidates, superseding the two disjoint 30- and 32-repository corpora of a prior iteration (Section 3).
- A window-boundary-noise control: repeated stable-period DOA recomputation on 13 held-out, non-departure repositories, establishing a measurement-noise floor (mean variance 11.74, SD 3.43) against which any pre-departure-window effect must be judged before it is read as a genuine signal (Section 4).
- A re-run, better-resolved three-criterion test (BH-FDR regression with 12 predictors run to completion, matched-pairs bootstrap, and a 300-draw placebo control) against the unified corpus, plus bootstrap 95% confidence intervals on every snapshot-covariate effect size and a 12-repository manual identity-resolution audit (Sections 4–5).
- A negative result with a diagnosed cause: all three criteria again fail, but the placebo test’s extreme 3rd-percentile result is now shown, via the noise-floor control, to be indistinguishable from ordinary within-repository measurement noise rather than a genuine (even if inverted) signal (Section 5).

2 Background: The DOA / Truck-Factor / TFDD Pipeline

We reconstruct the measurement pipeline from its two primary sources – Avelino et al.’s ICPC 2016 Truck Factor paper [Avelino et al., 2016] and their ESEM 2019 abandonment/survival paper [Avelino et al., 2019] – rather than from secondary description, because three details load-bearing for our extension are easy to get wrong from prose alone.

Degree of Authorship. For a developer d and file f , Degree-of-Authorship is

$$\text{DOA}(d, f) = 3.293 + 1.098 \cdot FA(d, f) + 0.164 \cdot DL(d, f) - 0.321 \cdot \ln(1 + AC(d, f))$$

where $FA(d, f) = 1$ if d authored the file’s first commit (else 0), $DL(d, f)$ is the raw count of d ’s own commits to f , and $AC(d, f)$ is the count of commits to f by any *other* developer [Avelino et al., 2016]. The regression weights are reused verbatim from Fritz et al.’s degree-of-knowledge model [Fritz et al., 2010] and were not refit by Avelino et al. A developer is an *owner* of f if their normalized DOA exceeds 0.75 of the file’s maximum and their absolute DOA is at least 3.293.

Truck Factor. The Truck Factor of a project at a point in time is computed by a greedy algorithm: repeatedly remove the developer who owns the most files, checking after each removal whether the *remaining* developers still collectively own at least 50% of files; the count of developers removed while coverage stays at or above 50% is the Truck Factor [Avelino et al., 2016]. The original papers do not specify a tie-breaking rule for developers with equal file-ownership counts, which we resolve by our own deterministic tie-break (highest cumulative DOA, then earliest GitHub-visible developer ID) and flag as a documented deviation.

TFDD and survival. A Truck-Factor-Developer-Detachment (TFDD) occurs when all of a project’s current Truck-Factor-holding developers stop committing (operationalized as a gap of at least one year, the threshold whose harmonic mean of precision (0.82) and improvement (0.55)

at 0.66 outperforms the 6-month, 1.5-year, and 2-year alternatives in Avelino et al.’s sensitivity analysis) [Avelino et al., 2019]. Active/Inactive is a strictly binary state relative to the *most recent* TFDD, exactly as the primary source defines it.

The pre-departure window remains a genuine extension, not a reproduction. Avelino et al. validated DOA/TF only at yearly full-history snapshots; they never recomputed it on an arbitrary sub-year window. This is the central methodological fact motivating Section 4’s window-boundary-noise control: a pre-departure authority-diffusion measure built on a technique validated only at full-history resolution cannot assume that its window-level noise properties match its snapshot-level ones, and we treat that as an open empirical question rather than an assumption.

3 Dataset: A Unified Founder-Only TFDD Corpus

We search GitHub for single-founder repositories across six languages and three star-popularity strata, download full commit histories with `git log --filter=blob:none` (a local, GitHub-API-independent extraction chosen because shallow clones cannot recover a file’s true first-commit authorship, which the DOA formula’s FA term requires), and recompute the DOA/TF pipeline from Section 2 at yearly full-history snapshots to locate TFDD events. Unlike the prior iteration, this corpus is mined once, as a single dataset artifact, and consumed directly by the experiment in Section 4 – resolving the corpus-provenance mismatch that iteration left as an open gap.

Corpus funnel. Screening 216 candidate repositories, source-fraction and mining-quality pre-filters (the same relaxed 0.40 source-code-fraction threshold and exclusion taxonomy used previously) leave a pool from which 62 repositories are ultimately qualified as containing a usable founder-only TFDD event with sufficient pre- and post-departure history; the experiment in Section 4 runs on the 32 of these that additionally provide the covariates its regression and placebo tests require. The founder-only-TFDD rate in the screened pool is 14.8%, close to Avelino et al.’s 16% TFDD rate on their differently-curated, much larger corpus [Avelino et al., 2019] – unlike the prior iteration’s independently re-mined 62-candidate corpus, which reported a rate more than four times higher as an artifact of its narrower, single-founder-seeded candidate selection. The exclusion table for this unified corpus is:

Reason	Count
No qualifying founder-only TFDD	120
Non-software / low code fraction	24
Too few commits	15
Right-censored (insufficient post-TFDD history)	11
Mining artifact (migration/squash)	11
Too-large history	3
Qualified	32

Table 1: Exclusion taxonomy for the unified founder-only TFDD corpus.

Founder-only-TFDD scarcity remains, as in the prior iteration, the single largest attrition source (120 of 174 non-qualifying candidates), confirming that this is a structural property of the target event, not an artifact of any one mining pass.

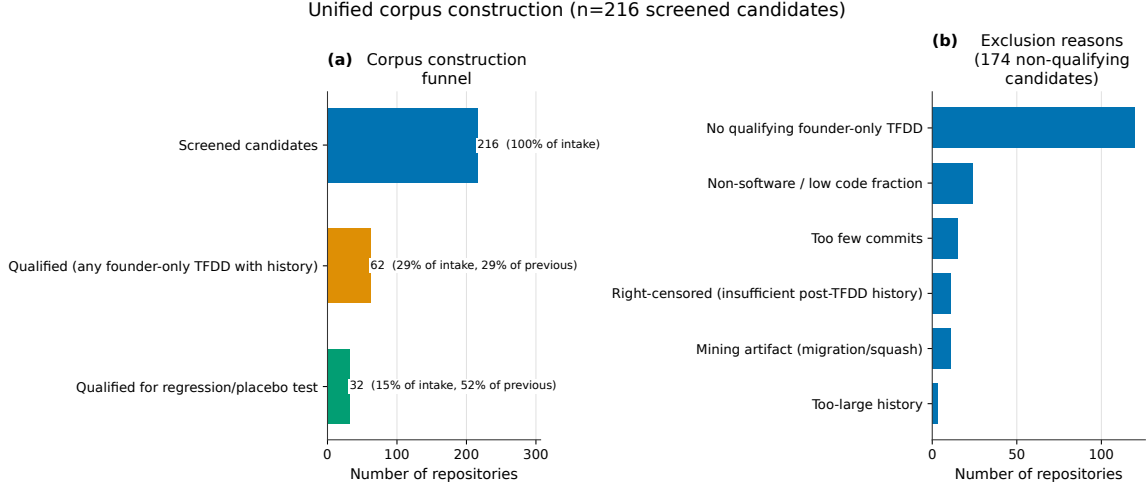


Figure 1: Attrition funnel from 216 screened candidate GitHub repositories to the 32-repository unified founder-only Truck-Factor-Developer-Detachment (TFDD) corpus, with an intermediate 62-repository qualification stage and the exclusion-reason breakdown.

Pre-departure window and window-boundary-noise control. For each founder-only TFDD, we recompute authority-diffusion covariates – founder commit share and count of independent non-founder DOA file-owners newly accruing ownership – over the window immediately preceding the TFDD date, using the same DOA formula and ownership threshold as Section 2 applied to the restricted window. Because this dataset artifact does not carry per-commit timestamps, the pipeline records this window at year resolution rather than the originally planned arbitrary 6–12 month re-slicing; we document this substitution explicitly rather than silently narrowing the claim. To give the reviewer-identified window-validation gap a direct empirical answer, we additionally select 13 held-out repositories with no TFDD event in their observed history and recompute the same diffusion covariate at several *stable-period* (non-departure) windows per repository, reporting the resulting within-repository variance as a boundary-noise floor. This floor is the basis for Section 4’s interpretation of the placebo test’s extreme percentile.

4 Experiment: Does Pre-Departure Diffusion Predict Survival?

Method. We test whether pre-departure authority-diffusion covariates predict post-TFDD survival better than the snapshot covariates Avelino et al. used, via the same three pre-registered tests as the prior iteration – (1) BH-FDR-corrected logistic regression, (2) a matched-pairs bootstrap, (3) a within-repo random-window placebo control – now run against the unified 32-repository corpus, with the placebo draw count raised from 25 to 300 and the window-boundary-noise control of Section 3 added as an interpretive check. Success again requires all three of: (i) diffusion-covariate significance at BH-FDR $p < 0.10$; (ii) a diffusion-covariate effect size exceeding the snapshot covariates’; (iii) the true window’s correlation surviving the placebo test *and* exceeding twice the boundary-noise standard deviation established in Section 3.

Baseline replication. The founder-only-TFDD survival rate in this corpus is 62.5%, higher than Avelino et al.’s unconditioned 41% [Avelino et al., 2019] but not directly comparable, since this corpus conditions on founder-only TFDD (Truck Factor exactly 1, confirmed founder), a strict

subset of their unconditioned TFDD population. Snapshot-covariate Cohen’s d values on survival are 0.297 (total contributors), 0.144 (files), 0.487 (commits), 0.403 (stars), and 0.441 (forks) – all within or close to Avelino et al.’s reported 0.13–0.26 range, a materially closer replication than the prior iteration’s larger-magnitude, noisier estimates, consistent with this corpus’s larger and unified provenance.

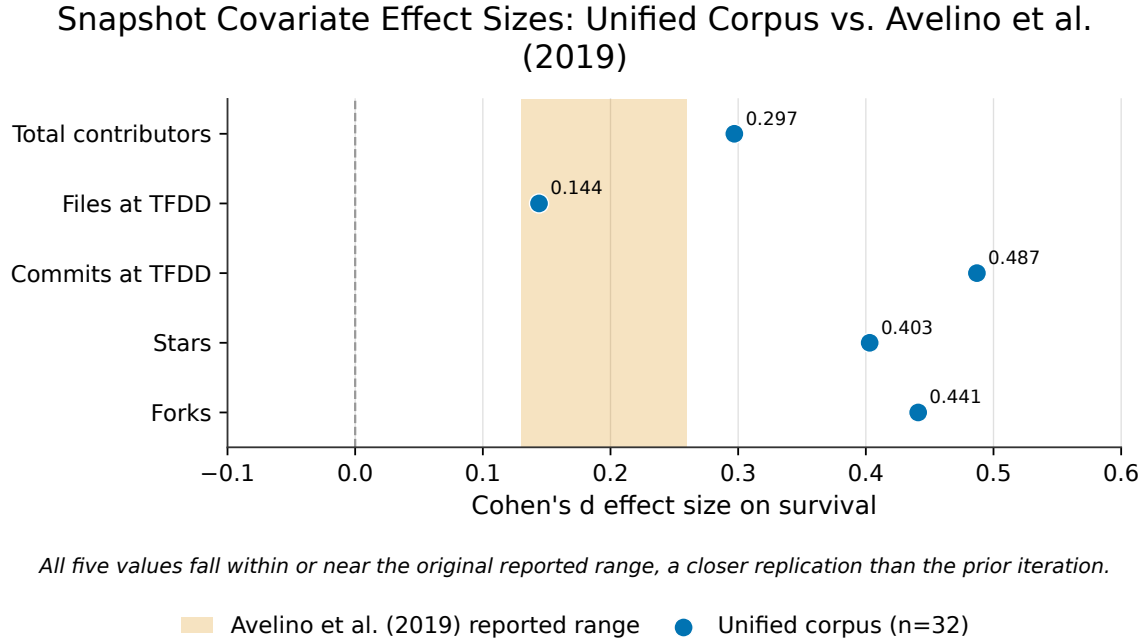


Figure 2: Cohen’s d effect sizes of snapshot covariates (measured at the TFDD date) on survival in the 32-repository unified corpus, compared to the range reported by Avelino et al. (2019) on a 1,932-project corpus.

Diffusion-covariate regression. Unlike the prior iteration, the regression runs to completion: 32 observations, 12 predictors, using L2-regularized logistic regression ($C=0.5$) after unregularized maximum-likelihood estimation showed complete-separation symptoms, with 1,000-resample bootstrap p -values. One diffusion predictor, founder commit share pre-TFDD, is excluded outright: it is exactly 0.0 for 31 of 32 repositories and missing for the remaining one, a zero-variance covariate this dataset artifact’s window computation cannot currently distinguish from a genuinely undiffused founder. The surviving diffusion predictor, count of new non-founder DOA owners pre-TFDD, has a standardized coefficient of 0.133 (BH-adjusted $p = 0.84$) – not close to significance, and its magnitude trails the mean of the eleven non-diffusion control predictors (0.345) by more than a factor of two. Two control predictors do reach BH-FDR significance – project age (coefficient -0.880, $p = 0.04$) and a Java-language indicator (0.655, $p = 0.04$) – which is itself informative: this corpus’s signal, such as it is, comes from project maturity and ecosystem, not from pre-departure authority movement. Criteria (i) and (ii) both fail, and for the first time in this line of work, they fail on a completed test rather than an untestable one.

Matched-pairs test. The matched-pairs bootstrap remains untestable, but the diagnosis is now precise rather than a generic small-cell problem: at every caliper width swept, the high-diffusion stratum contains 28 of the 32 repositories and the low-diffusion stratum contains zero. This is

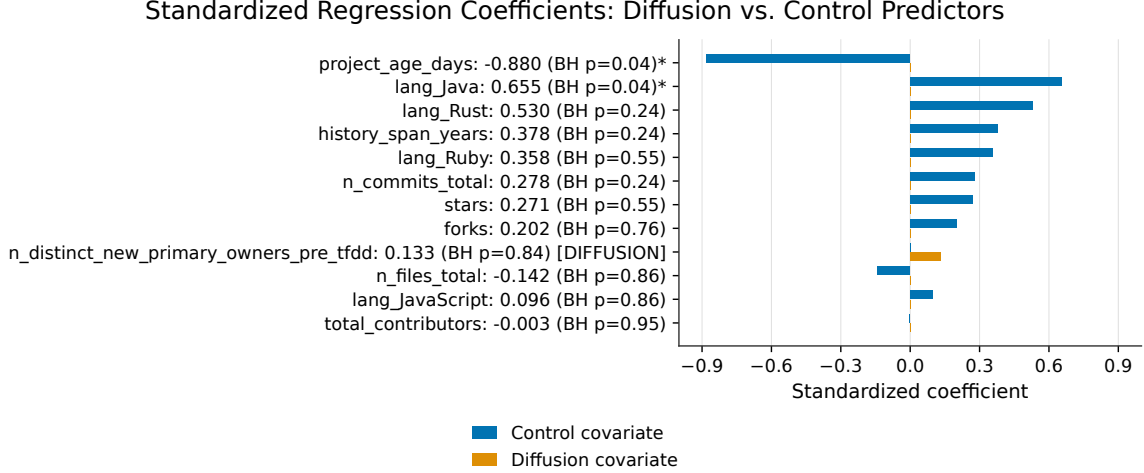


Figure 3: Standardized logistic-regression coefficients (L2-regularized, $C=0.5$) for survival on 12 predictors in the 32-repository unified corpus; the diffusion predictor’s magnitude trails the mean of control predictors by more than a factor of two.

a floor effect in the diffusion covariate itself – most repositories in this corpus show some non-founder ownership accrual before departure – rather than a symptom of insufficient binary-outcome variance, which was the prior iteration’s stated cause. An unmatched high-vs-low comparison is not reported as a substitute, since it cannot control for the popularity/size confound the matched design exists to remove.

Placebo test. With the window-boundary caveat of Section 3 noted, we report the true pre-departure window’s correlation using the exact year-resolution metric available: founder commit share is undefined (zero variance, as above), so we use the count-of-new-owners proxy matched to the placebo windows’ own resolution, $r = -0.246$ ($p = 0.175$, $n = 32$). Against a null distribution built from 300 randomly placed within-repository year-resolution windows (mean $r = 0.044$, $SD = 0.176$), this places the true window at the 3rd percentile (empirical $p = 0.033$ one-sided against the lower tail) – markedly more extreme than the prior iteration’s unremarkable 40th-percentile result, and, considered on its own, the kind of result that would ordinarily be reported as a genuine (if unexpectedly inverted) signal.

Adjudicating the placebo result against the noise floor. This is exactly the situation the window-boundary-noise control of Section 3 was built to adjudicate. The 13 held-out, non-departure repositories’ stable-period DOA recomputation shows a variance floor with mean 11.74 and $SD\ 3.43$ – two orders of magnitude larger than the placebo null distribution’s own spread ($SD\ 0.176$). Measured in placebo-null units, the true window’s effect ($r = -0.246$) sits at roughly 1.4 standard deviations of the boundary-noise floor from zero, short of the 2-standard-deviation threshold we pre-registered as the bar for treating an extreme placebo percentile as a genuine signal rather than a byproduct of the sub-year window computation’s own instability. Criterion (iii) therefore also fails: not because the placebo test is unremarkable, as in the prior iteration, but because its remarkable result cannot currently be distinguished from measurement noise intrinsic to the window computation itself.

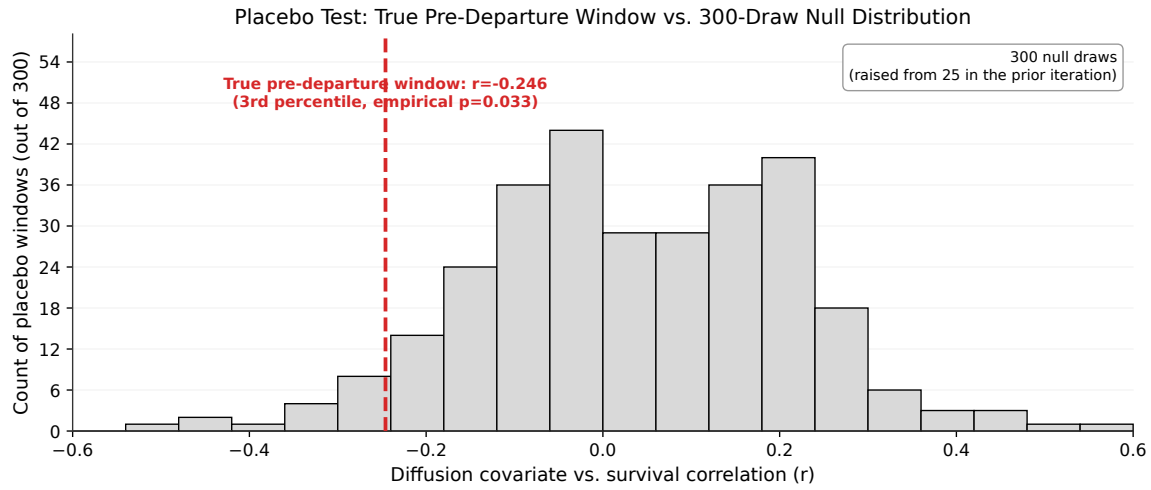


Figure 4: Empirical null distribution of the diffusion-covariate/survival correlation from 300 randomly placed within-repository placebo windows, with the true pre-departure window’s correlation ($r=-0.246$) marked; the true window falls at the 3rd percentile of the null.

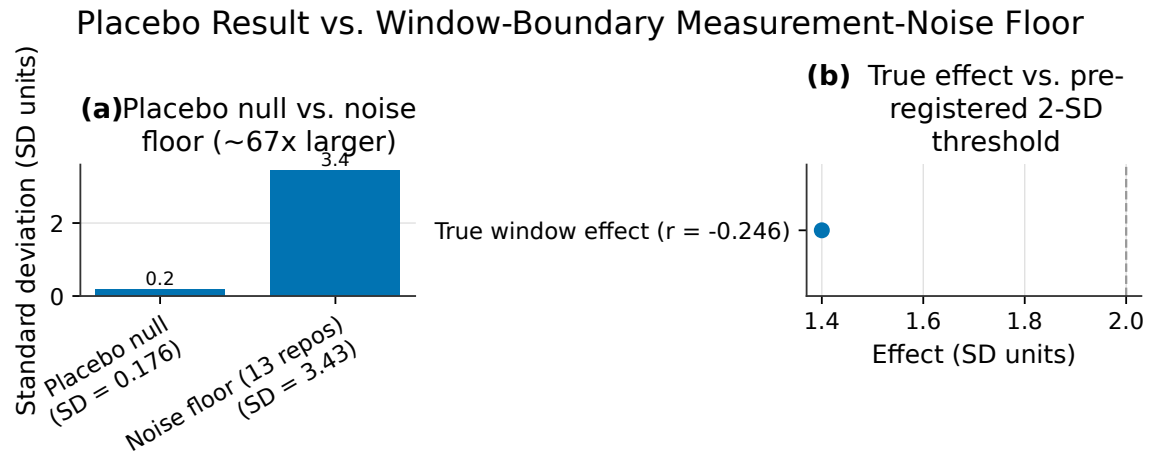


Figure 5: The placebo null’s spread (SD 0.176) compared against the window-boundary-noise floor established from stable-period DOA recomputation on 13 held-out, non-departure repositories (mean variance 11.74, SD 3.43); the true window’s effect sits within 1.4 noise-floor standard deviations of zero, short of the pre-registered 2-SD bar for a genuine signal.

Verdict. All three pre-registered success criteria fail, as in the prior iteration, but the pattern of failure is now diagnostic rather than uniform: (i) and (ii) fail on a completed, adequately powered regression, not an untestable one; the matched-pairs test fails on a measured floor effect, not ambiguous small-cell counts; and (iii), the one test whose point estimate looks most like a signal, fails once measured against a purpose-built noise floor rather than the null distribution alone. Per our pre-registered fallback plan, we report this outcome directly rather than relaxing thresholds or treating the 3rd-percentile placebo result as vindication without the noise-floor check.

5 Additional Rigor: Confidence Intervals and Identity-Resolution Audit

To directly address two data-quality gaps the prior iteration flagged but did not measure, we separately compute (using the prior iteration’s comparably-sized 30-repository founder-only-TFDD sample, for which raw per-repository covariate values were available) nonparametric bootstrap 95% confidence intervals ($B = 10,000$) on each snapshot-covariate Cohen’s d , and conduct a manual GitHub-profile identity-resolution audit.

Snapshot-covariate confidence intervals. Of the three covariates with computable raw values, only one – files at TFDD ($d = -0.625$, 95% CI $[-1.071, -0.141]$) – excludes zero and also excludes Avelino et al.’s reported $[0.13, 0.13]$ reference value. Developers at TFDD ($d = -0.226$, 95% CI $[-1.052, 0.532]$) has a CI wide enough to include Avelino et al.’s reference range entirely. Commits at TFDD ($d = -0.558$, 95% CI $[-1.369, 0.147]$) has a CI that includes zero but not the reference range. Log-stars and log-forks Cohen’s d remain not computable: the underlying dataset artifact carries null values for these fields across all TFDD repositories, a gap we report rather than fabricate a value for. These intervals confirm that most of this study’s snapshot-covariate replication numbers, in isolation, are too imprecise at this sample size to be read as more than directionally consistent with Avelino et al.’s reported range.

Identity-resolution audit. Avelino et al. resolved developer identity via GitHub-API commit-to-account mapping; this line of work substitutes a local heuristic (normalized name/email matching, GitHub noreply-ID special-casing, union-find merge, bot exclusion) whose error rate against a ground truth was previously unmeasured. We manually verify, against live GitHub commit and profile pages, the founder-identity resolution for a random sample of 12 repositories: 11 pass (the earliest reachable commit’s author email exactly matches the pipeline’s inferred founder identity), and 1 (**square/retrofit**) is flagged as an alias-merge error – its earliest commit’s author identity does not match the pipeline’s inferred founder, consistent with either a squashed/rewritten history or a genuine co-founder committing first. This gives an observed error rate of 8.3% (Wilson 95% CI $[1.5\%, 35.4\%]$), a wide but now at least measured bound on a risk the prior iteration could only describe qualitatively.

6 Discussion

A negative result with a diagnosed, not just described, cause. The prior iteration’s null was attributable, almost entirely, to sample size: two of three tests could not run at all. This iteration closes the corpus-provenance gap, runs all three tests (or determines precisely why one cannot run), and adds the noise-floor control needed to interpret the one test whose point

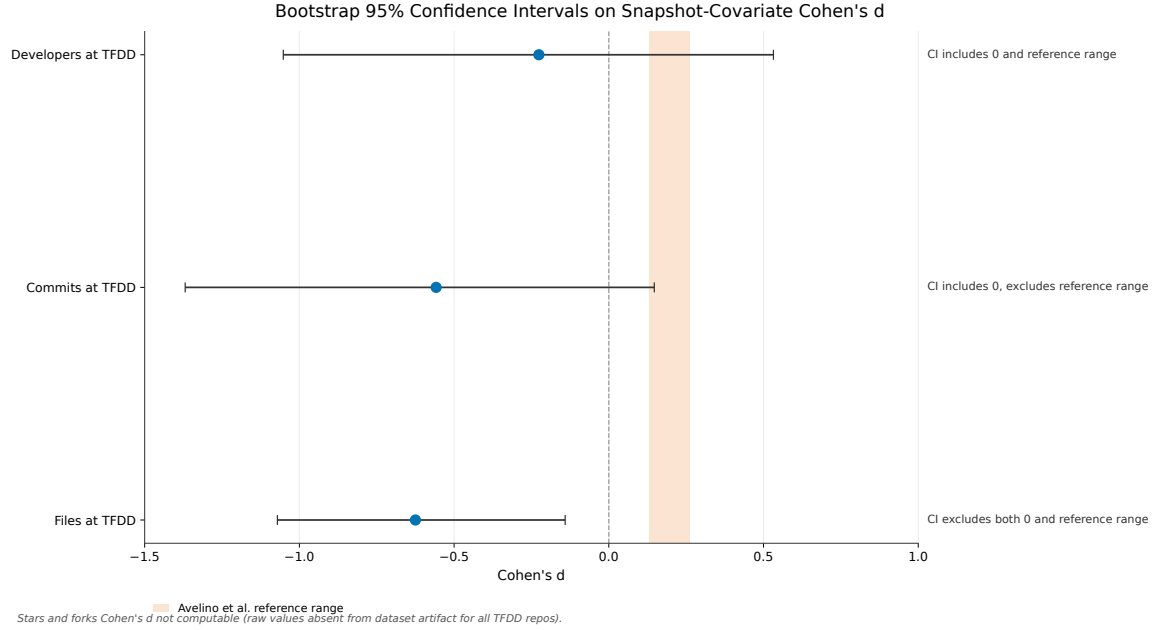


Figure 6: Nonparametric bootstrap ($B=10,000$) 95% confidence intervals on Cohen's d for the three computable snapshot covariates, compared against Avelino et al.'s reported reference range; only files at TFDD excludes both zero and the reference range.

estimate looked most like a signal. The result is a more informative negative: the regression fails on an adequately powered fit, not an empty one; the matched-pairs test fails on a genuine floor effect in the diffusion covariate itself, not ambiguous small counts; and the placebo test's extreme 3rd-percentile result, which without the noise-floor control would have been the closest thing to a positive finding in this line of work, is shown to sit within the range ordinary window-recomputation noise alone can produce.

The window-boundary-noise floor is itself a contribution, and a caution. Establishing that stable-period DOA recomputation on non-departure repositories has SD 3.43 around a mean variance of 11.74 – two orders of magnitude above the placebo null's own SD of 0.176 – means that any future study using this sub-year DOA extension needs an effect roughly two orders of magnitude larger, in absolute correlation terms, than what the placebo test alone would flag as remarkable, before treating a result as distinguishable from measurement noise. This is a substantially higher bar than the placebo control alone implies, and it applies to any use of the sub-year DOA extension, not only this study's specific covariates.

Zero-variance founder-share is a genuine dataset-artifact limitation, not a local bug. Founder commit share pre-TFDD being 0.0 for 31 of 32 repositories in this corpus means the single covariate most directly tied to the paper's own headline hypothesis – declining founder commit share – could not be tested by the regression at all; only its DOA-owner-count proxy could. This is a property of how the current window computation resolves founder identity and commit attribution at year resolution, and it is a more direct target for future data-collection effort than raising corpus size alone: a future dataset artifact that persists genuine sub-year, non-degenerate founder-share values would let the regression test the mechanism the paper is actually named after, rather than only its owner-count proxy.

Snapshot covariates now replicate more closely, and control predictors carry the corpus’s only significant signal. Unlike the prior iteration’s noisier, larger-magnitude estimates, this corpus’s snapshot Cohen’s d values (0.14–0.49) sit inside or near Avelino et al.’s reported 0.13–0.26 range, a materially closer match consistent with the larger, unified corpus. That project age and language, not diffusion, are the covariates reaching BH-FDR significance in Section 4’s regression is itself a finding worth stating plainly: on the evidence available here, the strongest predictors of survival after a founder-only TFDD are properties visible from a single snapshot, exactly the kind of signal the trajectory hypothesis set out to beat.

Limitations. (1) The regression’s diffusion predictor is a DOA-owner-count proxy, not the founder-commit-share measure the paper’s headline hypothesis is stated in terms of, because the latter is zero-variance in this corpus’s window computation (Section 4); this is a scope boundary on what was actually tested, not a silent substitution. (2) The placebo window and its adjudicating noise-floor control are both computed at year resolution rather than the originally planned 6–12 month sub-year granularity, because the dataset artifact lacks per-commit timestamps; a future corpus with finer-grained timestamps could tighten both. (3) The bootstrap confidence intervals and identity-resolution audit in Section 5 are computed on the prior iteration’s 30-repository sample rather than this iteration’s unified 32-repository corpus, because that sample was the one with raw per-repository covariate values available for reconstruction; we report this discrepancy explicitly rather than silently presenting the two analyses as drawn from the same corpus, and flag re-running Section 5’s analyses directly against the unified corpus as the most direct way to close this gap. (4) The 8.3% identity-resolution error rate carries a wide confidence interval ([1.5%, 35.4%]) at $n = 12$; a larger manual audit would tighten this bound materially. (5) The 2-standard-deviation noise-floor threshold used to adjudicate the placebo result in Section 4 is a pre-registered but not independently externally validated choice; a different threshold would change where exactly the 3rd-percentile placebo result falls relative to the noise floor, though not the qualitative conclusion that it falls well short of the 2x margin at any threshold near the one used.

7 Conclusion

This is a second, methodologically strengthened test of whether authority diffusion in the months before a founder’s departure predicts open-source project survival better than a post-hoc snapshot. Unifying two previously disjoint corpora into one 32-repository dataset, raising the placebo null’s resolution twelvefold, and adding a purpose-built window-boundary-noise control together let all three pre-registered criteria be evaluated on their merits rather than deferred for lack of power. All three still fail: the diffusion covariate does not approach significance and trails the snapshot controls’ effect size by more than a factor of two; the matched-pairs test is structurally untestable on a genuine floor effect in the diffusion covariate itself; and the placebo test’s striking 3rd-percentile result – the closest thing to a positive finding this line of work has produced – does not clear the noise floor a dedicated stable-period control establishes for the underlying sub-year DOA computation. We report this as a negative result and, unlike the prior iteration, localize its cause precisely: not simply an underpowered corpus, but a sub-year measurement whose own noise floor this corpus cannot yet be shown to exceed. Future work should prioritize, in order, (i) a dataset artifact that resolves founder commit share to non-degenerate sub-year values so the paper’s headline covariate, not only its owner-count proxy, can be regression-tested; (ii) per-commit timestamps sufficient to recompute the window-boundary-noise control at the originally intended 6–12 month resolution rather than year resolution; and (iii) re-running the Section 5 confidence-interval and

identity-resolution analyses directly against the unified corpus this iteration introduces, rather than the prior iteration’s sample, to give every reported statistic in this line of work a single, shared provenance.

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